

Isolation of Protease-Producing Bacteria and Optimization of Protease Production Using Box-Behnken Design

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Abstract

Proteases are important hydrolytic enzymes with broad applications in pharmaceutical, medical, food, detergent, leather, and other industrial processes. The present study focused on the isolation, screening, and optimization of protease-producing bacterial strains for enhanced enzyme production. Soil samples were collected and screened for proteolytic activity using gelatin agar as a selective screening medium. Among the bacterial isolates obtained, three promising strains, designated DK1, DK2, and DK3, demonstrated significant proteolytic activity based on clear-zone formation. Protease production was further evaluated under different cultivation conditions to identify factors influencing enzyme yield. Optimization studies were performed by investigating key parameters, including pH, temperature, and incubation time. A Box–Behnken experimental design was applied to evaluate the combined effects of these variables and determine conditions that favored maximum protease production. The optimized conditions resulted in improved enzyme production compared with the initial cultivation conditions. The findings demonstrate the potential of the selected bacterial isolates as sources of extracellular protease and provide useful information for optimizing microbial enzyme production. The study may contribute to the development of efficient and economical approaches for producing proteases with potential applications in pharmaceutical and other industrial processes.

Keywords: protease; bacteria; soil isolates; casein-TCA assay; Box-Behnken design; response surface methodology; enzyme optimization

1. Introduction

Proteases are enzymes that catalyze the hydrolysis of peptide bonds and convert proteins into smaller peptides and amino acids. Because of this function, proteases participate in many biological processes and are also among the most commercially important groups of enzymes. Microbial proteases are particularly attractive for biotechnology because microorganisms can be cultivated rapidly, require relatively limited space, can be manipulated genetically, and provide broad biochemical diversity. (Kumari *et al.*, 2012). Bacterial proteases have therefore been widely investigated for industrial applications, especially in food processing, detergents, leather treatment, pharmaceuticals, chemical processing, textiles, and management of protein-rich wastes. (Gupta, *et al.*; 2002a; Genckal, 2004; Ikram, 2008; Nadeem, 2009; Ray, 2012).

Protease production is influenced by environmental and culture variables such as pH, temperature, and incubation time. These factors may interact with one another, so changing a single factor at a time does not always reveal the most favorable combination of conditions. Response surface methodology offers a statistical approach for evaluating several factors simultaneously. The Box-Behnken design (BBD), in particular, can be used to fit a quadratic model, assess factor interactions, and estimate conditions that maximize a measured response.

The objective of this study was to isolate protease-producing bacteria from soil, screen the isolates for proteolytic activity, characterize the most active isolates by Gram staining and colony morphology,

assess amylase production, and optimize protease production using a three-factor Box-Behnken design involving pH, temperature, and incubation time.

2. Materials and Methods

2.1 Sample collection and isolation of protease-producing bacteria

One gram of each collected sample was suspended with nine ml of the normal saline (0.85% NaCl). From the sample, 1 ml of supernatant was serially diluted up to 10^{-6} dilution separately using sterile normal saline solution.

From the each of the dilution 10^{-3} to 10^{-6} , 100 μ l was drawn and spread on the Luria –Bertani (LB) agar (HiMedia) medium with 1% Gelatin. The plates were incubated at 37°C for 24 hr, after 24 hr different types of bacterial colonies will be seen. Add acidic HgCl₂ in the plates, clear zone of gelatin hydrolysis was noted for each plate. After measuring the mean diameter of clear zone of hydrolysis, top three positive isolates with wider zone were taken for further studies.

2.2 Tyrosine standard curve and Protease assay

To prepare the standard curve of tyrosine, there are used 0.5M Na₂CO₃, 50mM of Sodium phosphate buffer, diluted 1N Folin reagent and 10 mg/ml of tyrosine stock solution were used. A required amount of buffer and tyrosine were added in each tube except blank. Then 2.5 ml of Na₂CO₃ was added in each tubes including blank. After these add 0.5 ml of 1N Folin reagent in each tubes including blank, the solution was mixed immediately and kept for 30 min at room temperature. Finally, the Optical Density (OD) was measured at 660nm using spectrophotometer and the standard Curve was plotted..

For the casein-TCA protease assay, The selected isolates were inoculated into the LB Broth medium and incubate at 37°C. The culture broth was centrifuged at 10000 rpm for 15 min at 4°C. Protease activity was determining using casein as a substrate. The reaction mixture contained total volume of 2 ml which in turn composed of 1ml of 1% casein in 50mM sodium phosphate buffer and 1 ml enzyme solution. After the 20 min of incubation at 37°C add 2 ml of 10% TCA (trichloroacetic acid) and again incubate at 37°C for 20 min. after the incubation the solution will be transferred in the falcon tubes and centrifuge at 7000 rpm for 20 min. 0.5 ml of clear supernant was mixed with the 2.5 ml of 0.5M Na₂CO₃ and 0.5 ml of 1N Folin reagent. After incubation of 20 min at 37°C, absorbance was measured at 660nm against a reagent blank. The process was carried out in triplicates. One unit of protease activity was defined as amount of enzyme that releases 1 μ g amino acid equivalent to tyrosine per min under the standard assay conditions (Hema and Shiny, 2012, Sevinc and Demirkan, 2011).

2.3 Gram staining, colony morphology, and amylase test

The three selected isolates were examined by Gram staining. Smears were air-dried, heat-fixed, stained with crystal violet, treated with Gram's iodine, decolorized with 95% ethanol, and counterstained with safranin. Slides were examined microscopically under the 100 $\times$ objective. Purple cells were recorded as Gram-positive and pink or red cells as Gram-negative.

Colony characteristics were observed from fresh cultures grown on LB agar. Pigmentation, form, margin, elevation, texture, and size were recorded..

Amylase test

The isolated top 3 strains were overnight grown and pinch upon the starch agar plate (LB agar + 2% starch). The plates were then incubated for 24hr and flooded with freshly prepared iodine solution. Formation of zone of clearance around the colonies was observed

Box-Behnken experimental design and statistical analysis

A three-factor, three-level Box-Behnken design was used to investigate the effects of medium pH, temperature, and incubation time on protease activity. The central values were pH 7, 37°C, and 24 h. Protease activity (U/ml) was treated as the response variable, and a second-order quadratic polynomial model was fitted. Analysis of variance (ANOVA) and the F-test were used to evaluate model significance at a 95% confidence level. Model quality was assessed by the coefficient of determination (R^2). Three-dimensional response surfaces and two-dimensional contour plots were generated using Design-Expert software, and optimum conditions were estimated through the software's desirability function and response-surface analysis.

Table 1. BBD and Response result

Runs	Factors	Factors	Factors	Response: Protease activity(U/ml)	Response: Protease activity(U/ml)	Response: Protease activity(U/ml)
Runs	X1	X2	X3	DK1	DK2	DK3
1	0	0	0	14.232	11.919	15.552
2	0	0	0	16.213	11.886	15.734
3	-1	-1	0	12.695	8.495	10.367
4	0	-1	-1	12.547	12.398	12.728
5	0	1	1	8.947	10.631	13.571
6	-1	0	1	9.012	10.367	13.240
7	0	0	0	15.157	12.068	16.180
8	0	1	-1	13.372	13.471	13.339
9	-1	1	0	11.160	12.613	12.613
10	0	0	0	14.430	13.455	14.793
11	0	-1	1	9.871	12.563	11.225
12	-1	0	-1	11.869	14.066	13.174
13	0	0	0	14.480	12.778	14.607
14	1	0	-1	14.281	8.368	12.447
15	1	1	0	11.704	10.928	12.431
16	1	0	1	10.053	11.638	10.928
17	1	-1	0	11.671	10.994	8.319

Table 2. Factors and levels used in the Box-Behnken design

Variable	-1	0	+1
pH	5	7	9
Temperature (°C)	30	37	44
Incubation time (h)	24	72	120

3. Results

3.1 Isolation and screening of protease-producing bacteria

Fifty bacterial isolates showing gelatin hydrolysis were recovered from the soil samples. Based on the size of the clearance zone, three isolates were selected for detailed study and designated DK1, DK2, and DK3. Their zones of inhibition were 13 mm, 12 mm, and 14 mm, respectively, with DK3 producing the largest zone. The gelatin-hydrolysis plate and the distribution of clearance zones are shown in Figures 1 and 2.



Figure 1. Formation of a clearance zone on gelatin agar indicating protease production.

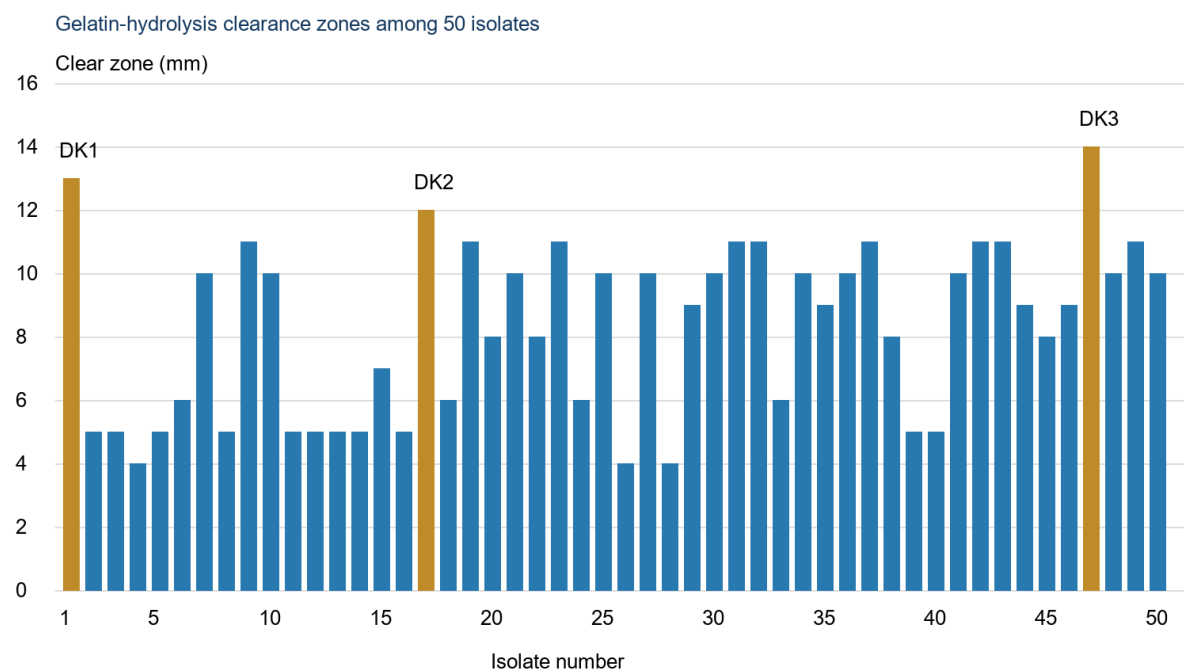


Figure 2. Distribution of gelatin-hydrolysis clearance zones among the 50 isolates. The selected isolates corresponding to DK1, DK2, and DK3 are annotated.

Table 3. Screening and initial protease activity of the three selected isolates

Isolate	Gelatin clearance zone (mm)	Protease activity (U/ml)
DK1	13	28.662
DK2	12	29.062
DK3	14	29.672

3.2 Tyrosine standard curve and protease activity

The tyrosine standard curve showed a strong linear relationship, with an R^2 value of 0.9963. Using the casein-TCA assay and this standard curve, the initial protease activities of DK1, DK2, and DK3 were calculated as 28.662, 29.062, and 29.672 U/ml, respectively. DK3 therefore showed the highest measured protease activity among the three selected isolates. The standard curve and the comparative protease activities are presented in Figures 3 and 4.

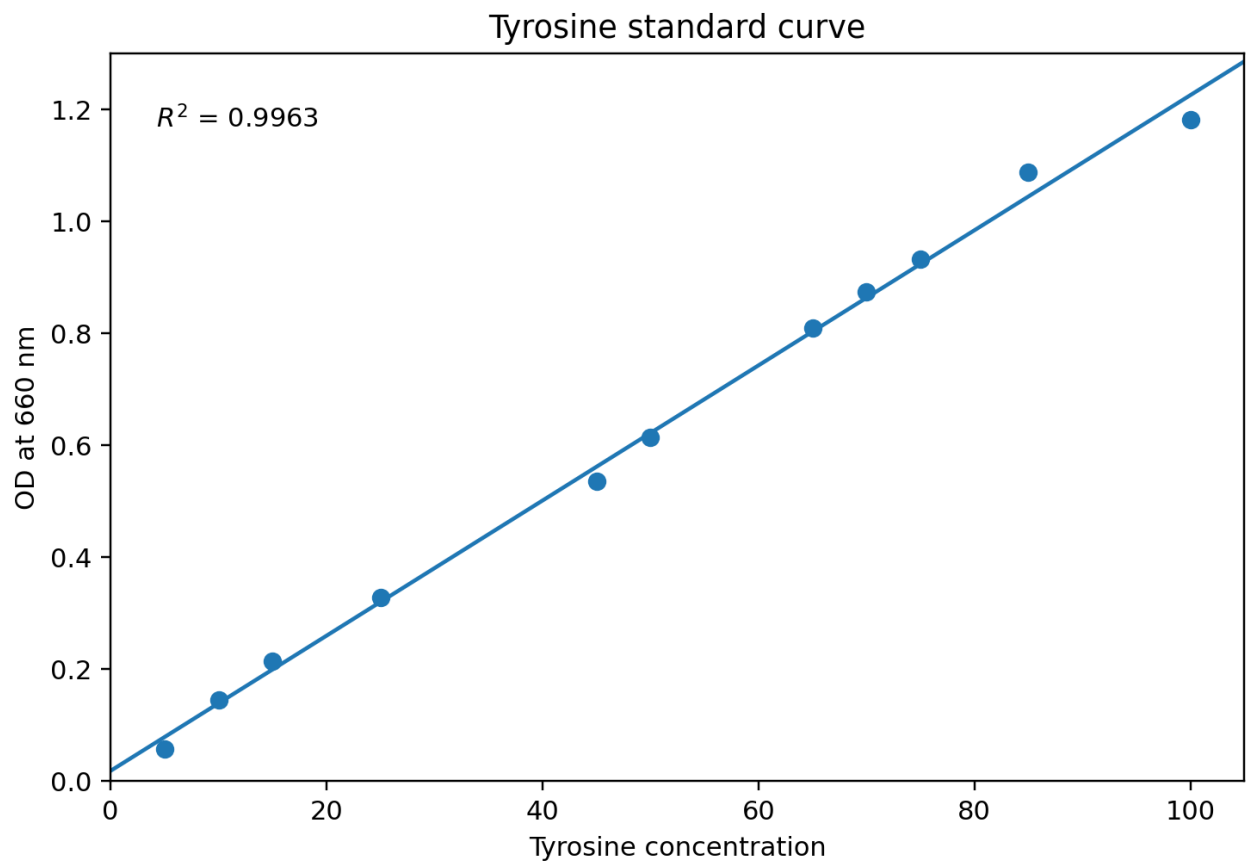


Figure 3. Tyrosine standard curve used for protease-activity determination ($R^2 = 0.9963$).

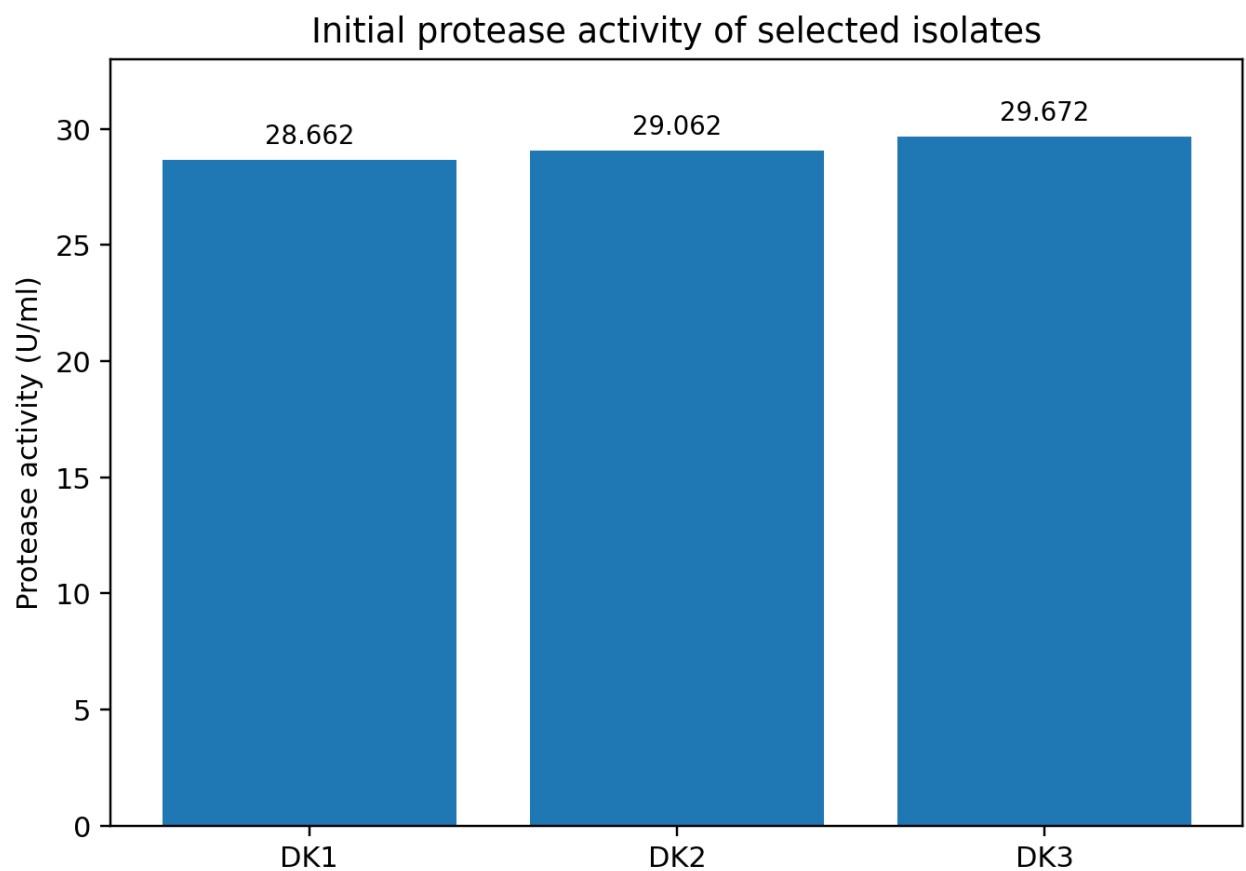


Figure 4. Initial protease activities of DK1, DK2, and DK3 measured by the casein-TCA assay.

3.3 Phenotypic characterization and amylase activity

Gram staining showed that DK1 and DK3 retained the primary crystal violet stain and were Gram-positive, whereas DK2 appeared pink after counterstaining and was Gram-negative. All three isolates formed large colonies. DK1 was white, circular, entire, raised, and butyrous; DK2 was off-white, circular, entire, convex, and viscid; and DK3 was buff-colored, irregular, entire, flat, and viscid. In the starch hydrolysis test, clear zones were observed for DK2 and DK3, indicating that these two isolates also produced amylase. Representative Gram-stained fields and the amylase-screening plate are shown in Figures 5 and 6.

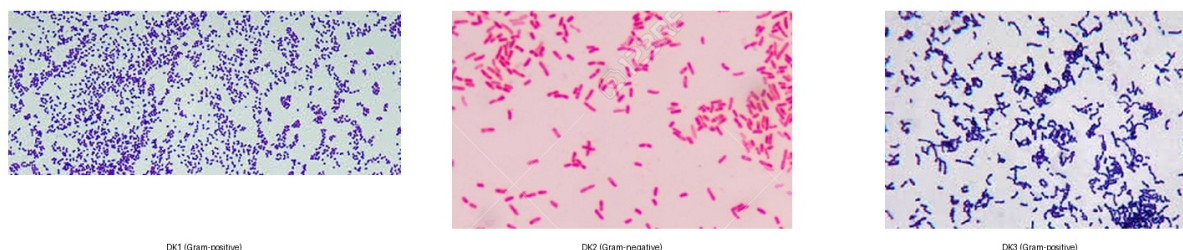


Figure 5. Gram-staining images of DK1, DK2, and DK3. DK1 and DK3 were Gram-positive, whereas DK2 was Gram-negative.

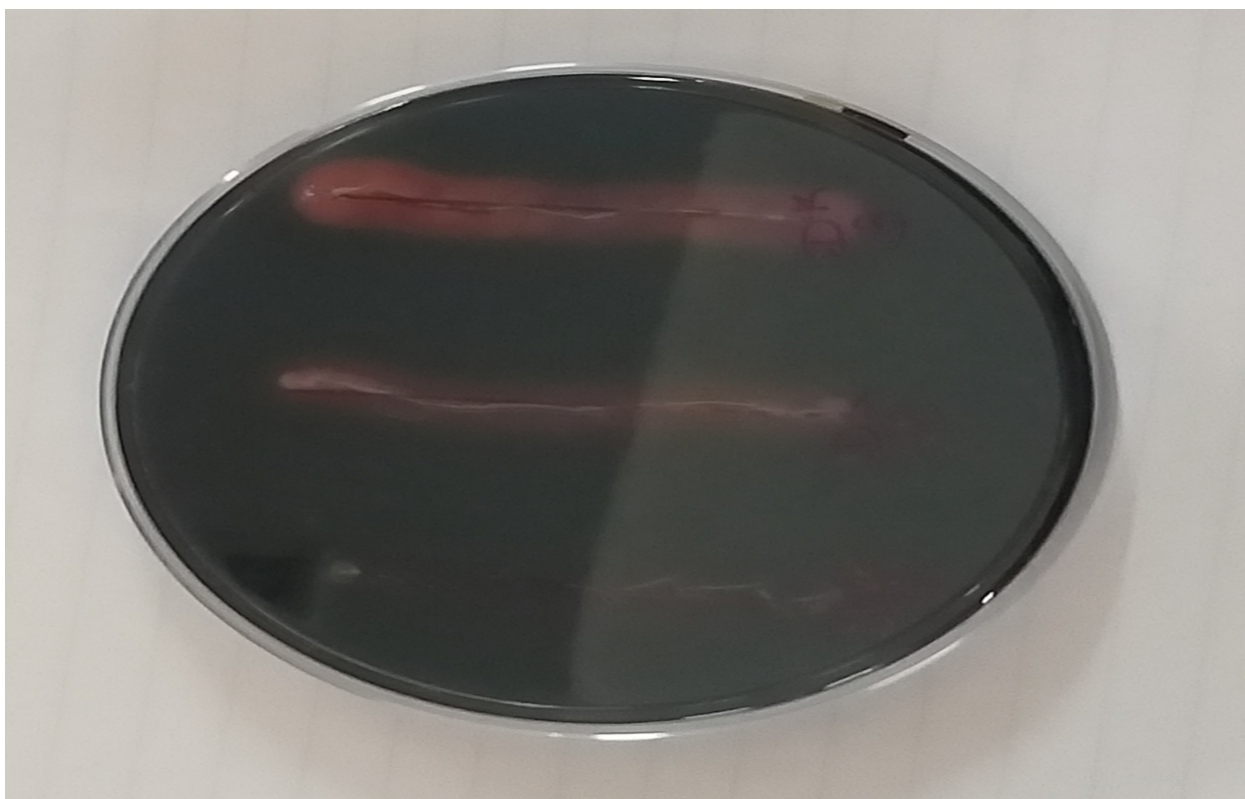


Figure 6. Amylase-screening plate. Clear zones were observed for DK2 and DK3 after iodine treatment.

Table 4. Morphological characteristics of selected protease-producing isolates

Characteristic	DK1	DK2	DK3
Pigmentation	White	Off-white	Buff
Form	Circular	Circular	Irregular
Margin	Entire	Entire	Entire
Elevation	Raised	Convex	Flat
Texture	Butyrous	Viscid	Viscid
Size	Large	Large	Large

3.4 Optimization by Box-Behnken design

The Box-Behnken experiments demonstrated that protease activity varied with pH, temperature, and incubation time and that the factors interacted with one another. The fitted quadratic models were statistically significant for all three isolates. For DK1, the regression model had an F value of 11.61, $p = 0.0019$, and $R^2 = 0.9372$. For DK2, the F value was 7.52, $p = 0.0072$, and $R^2 = 0.9063$. For DK3, the model produced an F value of 15.31, $p = 0.0008$, and $R^2 = 0.9513$. These R^2 values indicate that the models accounted for a high proportion of the observed variation in protease activity.

Table 5. ANOVA statistics for the Box-Behnken models

Isolate	Source	SS	df	Mean square	F	p value / R^2
DK1	Regression	71.82	9	7.98	11.61	0.0019 / 0.9372
DK1	Residual	4.81	7	0.68	—	—
DK2	Regression	40.80	9	4.53	7.52	0.0072 / 0.9063
DK2	Residual	4.22	7	0.6026	—	—
DK3	Regression	64.15	9	7.13	15.31	0.0008 / 0.9513
DK3	Residual	3.26	7	0.4655	—	—

Response-surface analysis indicated isolate-specific regions of high protease activity. For DK1, the favorable region was approximately pH 7.0–7.4 and 36–40°C, with an incubation period around 50–55 h. For DK2, the favorable region was pH 5.8–6.0 and 43–45°C, with approximately 24–43 h incubation. DK3 showed a favorable region around pH 6.5–7.0 and 35–40°C, with approximately 57–62 h incubation. The optimized point estimates generated by the design were 36°C, pH 7.2, and 51 h for DK1; 43°C, pH 5.9, and 24 h for DK2; and 38°C, pH 6.7, and 62 h for DK3. The corresponding Box-Behnken contour and three-dimensional response-surface plots are shown in Figures 7–9.

Table 5. Optimized protease-production conditions predicted by Box-Behnken design

Isolate	Temperature (°C)	pH	Incubation time (h)
DK1	36	7.2	51
DK2	43	5.9	24
DK3	38	6.7	62

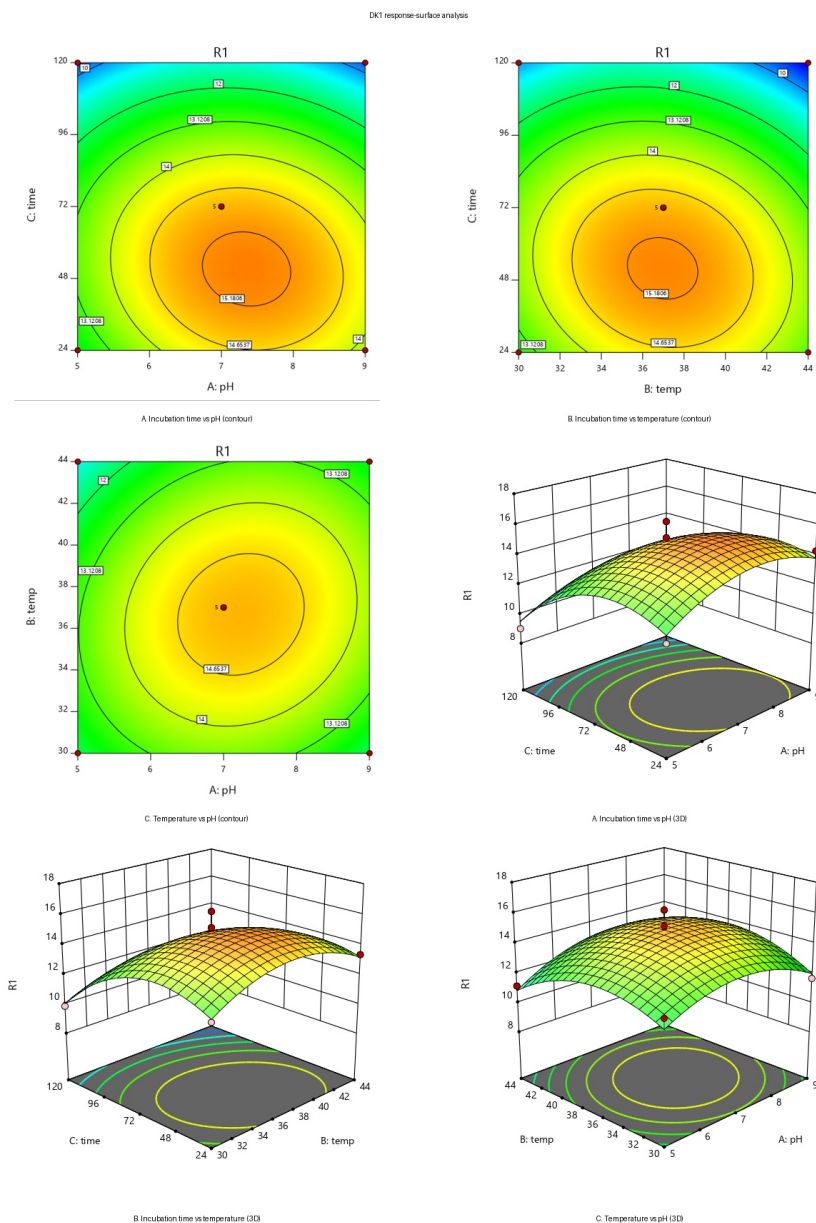


Figure 7. DK1 Box-Behnken response-surface plots showing the effects and interactions of pH, temperature, and incubation time on protease activity.

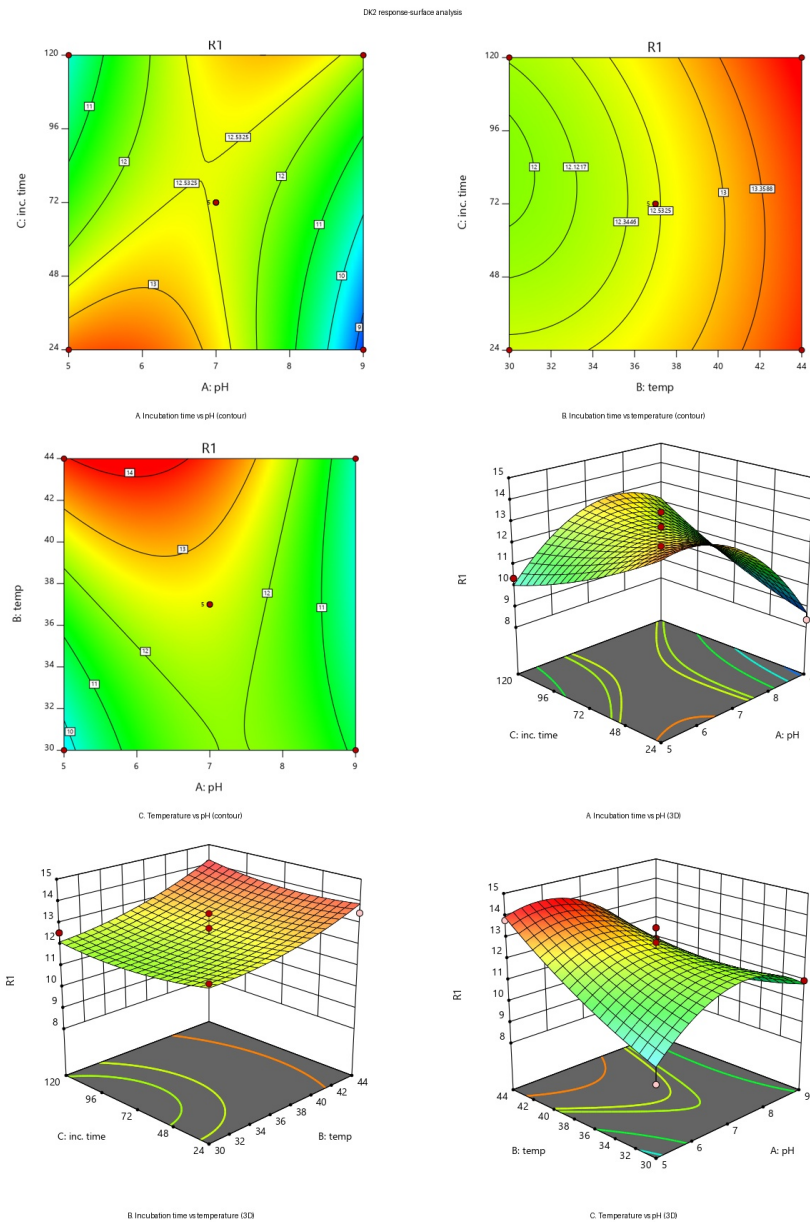


Figure 8. DK2 Box-Behnken response-surface plots showing the effects and interactions of pH, temperature, and incubation time on protease activity.

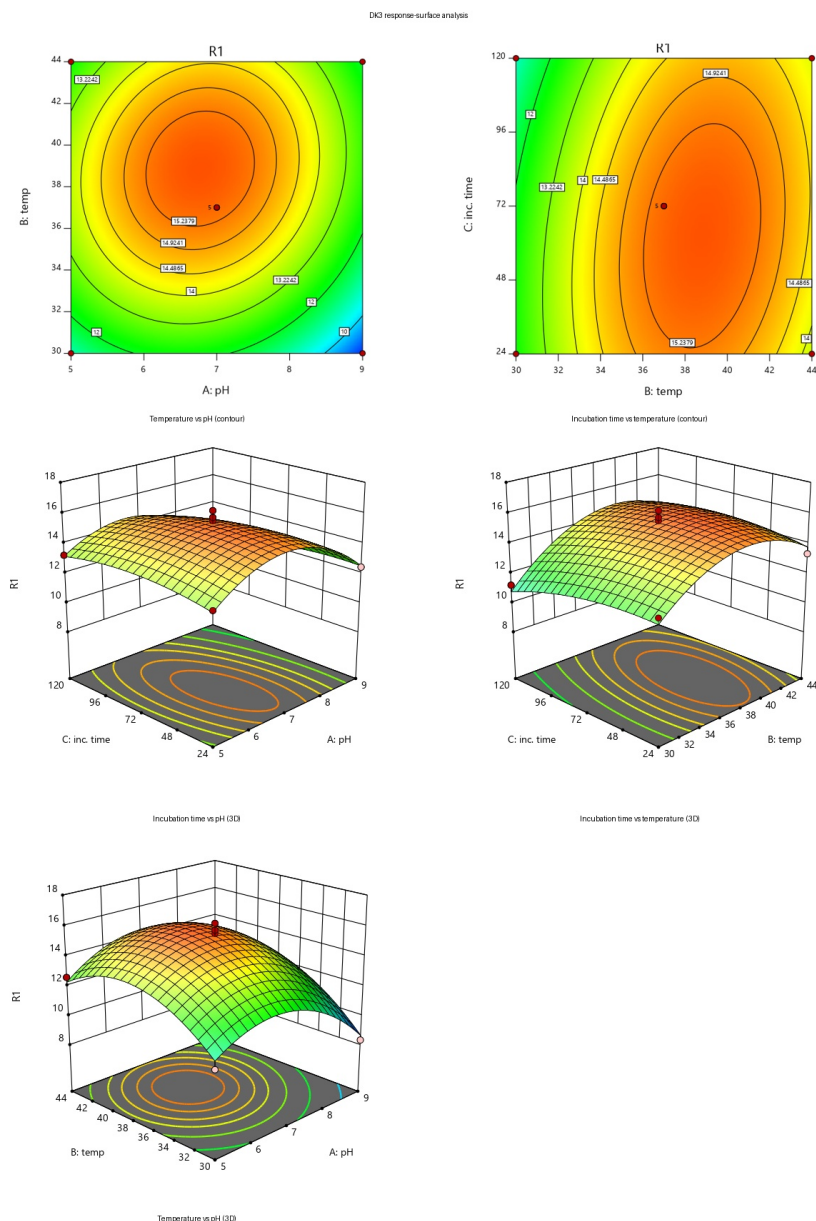


Figure 9. DK3 Box-Behnken response-surface plots showing the effects and interactions of pH, temperature, and incubation time on protease activity.

4. Discussion

The screening approach based on gelatin hydrolysis successfully identified a comparatively small subset of high-performing isolates from a larger soil bacterial population. Selection of DK1, DK2, and DK3 on the basis of clear-zone formation was supported by quantitative protease assays, in which all three isolates produced measurable extracellular protease activity close to 29 U/ml. DK3 had both the largest gelatin clearance zone and the highest initial protease activity, suggesting that the plate-screening result was useful for identifying a strong candidate for further optimization.

The phenotypic results also showed that protease production was not limited to a single Gram-reaction type. DK1 and DK3 were Gram-positive, while DK2 was Gram-negative. The positive amylase tests for DK2 and DK3 indicate that these isolates may have broader hydrolytic capabilities in addition to protease production. Such multi-enzyme activity can be relevant in biological processing systems in which mixtures of proteins and carbohydrates are present. However, the present study measured amylase only qualitatively by starch clearance, so the magnitude and industrial relevance of amylase production remain to be quantified.

The optimization study showed that the best production conditions differed markedly among isolates. DK1 favored near-neutral conditions, DK2 favored a mildly acidic pH combined with the highest temperature among the three optima, and DK3 favored a slightly acidic-to-neutral pH with an intermediate temperature. Incubation time also differed substantially, from 24 h for DK2 to 62 h for DK3. These differences demonstrate why isolate-specific optimization is necessary rather than assuming that one set of culture conditions is suitable for all protease-producing bacteria.

The statistical models were significant and explained approximately 91–95% of the variability in protease activity. The highest R^2 and F value were obtained for DK3, supporting a particularly strong fit of the quadratic model for that isolate. The curved response surfaces reported in the thesis indicate interactive effects among pH, temperature, and incubation time. Thus, the Box-Behnken design provided more information than a sequence of single-factor experiments by simultaneously describing the combined effects of the production variables.

From an application perspective, proteases are relevant to many industries because they hydrolyze protein substrates under controlled conditions. Microbial proteases in food and feed processing, dairy and baking, detergent formulations, leather treatment, pharmaceutical applications, silver recovery from photographic films, chemical synthesis, textiles, and treatment of proteinaceous waste. The three isolates studied here therefore represent potential starting material for further characterization. Before industrial use, however, additional work would be required to identify the organisms, confirm the optimized conditions experimentally, determine enzyme stability and substrate specificity, evaluate production scale-up, and assess safety.

5. Conclusion

Protease-producing bacteria were successfully isolated from soil, and three isolates—DK1, DK2, and DK3—were selected on the basis of gelatin hydrolysis and protease activity. DK3 showed the highest initial protease activity (29.672 U/ml), while DK1 and DK3 were Gram-positive and DK2 was Gram-negative. DK2 and DK3 also produced amylase. Box-Behnken response surface analysis demonstrated significant effects and interactions of pH, temperature, and incubation time on protease production. The optimized conditions were 36°C, pH 7.2, and 51 h for DK1; 43°C, pH 5.9, and 24 h for DK2; and 38°C, pH 6.7, and 62 h for DK3. The significant regression models and high R^2 values support the usefulness of the Box-Behnken approach for optimizing protease production.

References

- Sumantha, A., Larroche, C. and Pandey, A., 2006. Microbiology and industrial biotechnology of food-grade proteases: a perspective. *Food Technology and Biotechnology*, 44(2), p.211.
- Kaur, I., 2011. BIOCHEMICAL AND MOLECULAR CHARACTERIZATION OF ALKALINE PROTEASE OF BACILLUS CIRCULANS MTCC 7906 (Doctoral dissertation, PAU Ludhiana).
- Rao, M.B., Tanksale, A.M., Ghatge, M.S. and Deshpande, V.V., 1998. Molecular and biotechnological aspects of microbial proteases. *Microbiol. Mol. Biol. Rev.*, 62(3), pp.597-635.
- Anduaem, B. and Miesho, H., 2015. *Protease Production from Bacterial Isolates of Traditional Leather Processing Ponds found Around Gondar Town* (Doctoral dissertation, Haramaya University).
- Kebede, A. and Yimer, D., 2014. *Production and Characterization of Bacterial Protease from Isolates of Soil and Agro-Industrial Wastes* (Doctoral dissertation, Haramaya University).
- Deák, T. and Farkas, J., 2013. *Microbiology of thermally preserved foods: canning and novel physical methods*. DEStech Publications, Inc.
- Rao, M.B., Tanksale, A.M., Ghatge, M.S. and Deshpande, V.V., 1998. Molecular and biotechnological aspects of microbial proteases. *Microbiol. Mol. Biol. Rev.*, 62(3), pp.597-635.

- Mehta, A., 2010. Microbial proteases and their applications. *Industrial Exploitation of Microorganisms*. Maheshwari DK, Dubey RC, Sarvanamurthu R (eds). IK International Publishing House Pvt. Ltd., New Delhi, India, pp.199-226.
- Hamza, T.A. and Azmach, N.N., Studies on Protease Producing Bacterial Biodiversity from Soil Collected from Arba Minch University, Abaya Campus, Southern Ethiopia.
- Jewell, S.N., 2000. *Purification and characterization of a novel protease from Burkholderia strain 2.2 N* (Doctoral dissertation, Virginia Tech).
- Jogdand, K., Surywanshi, D., Munde, T. and Kate, S., Screening Of Protease Producer from Soil & Its Application As A Biocleaner.
- Khan, M.A., Ahmad, N., Zafar, A.U., Nasir, I.A. and Qadir, M.A., 2011. Isolation and screening of alkaline protease producing bacteria and physio-chemical characterization of the enzyme. *African Journal of Biotechnology*, 10(33), pp.6203-6212.
- Rupali, D., 2015. Screening and isolation of protease producing bacteria from soil collected from different areas of Burhanpur Region (MP) India. *Int. J. Curr. Microbiol. Appl. Sci*, 4, pp.597-606.
- VG, V.P., Preethi, S., Karthikeyan, S. and NG, R.B., 2016. ISOLATION AND IDENTIFICATION OF PROTEASE PRODUCING BACTERIA FROM SOIL.
- Hamza, T.A. and Letebo, A., Isolation and Screening of Protease Producing Bacteria from Local Habitat for Polyester Recovery from Waste Photographic Films.
- Manorma, K., Sharma, S., Singla, H., Kaundal, K. and Kaur, M., 2017. Screening and Isolation of Protease Producing Bacteria from Rhizospheric Soil of Apple Orchards from Shimla District (HP), India. *Int. J. Curr. Microbiol. App. Sci*, 6(5), pp.249-255.
- Vijayalakshmi, T.M. and Murali, R., 2015. Isolation and screening of *Bacillus subtilis* isolated from the dairy effluent for the production of protease. *International Journal of Current Microbiology and Applied Sciences*, 4(12), pp.820-827.
- Pokhrel, B., Pandeya, A., Gurung, S., Bista, G., Kandel, S., Kandel, R.C. and Magar, R.T., 2014. Screening and optimization of extra cellular protease from bacteria isolated from sewage. *European Journal of Biotechnology and Bioscience*, 2(1), pp.46-49.
- Sharma, A.K., Sharma, V., Saxena, J., Yadav, B., Alam, A. and Prakash, A., 2015. Isolation and screening of extracellular protease enzyme from bacterial and fungal isolates of soil. *Int J Sci Res Environ Sci*, 3, pp.0334-0340.
- Malathi, S. and Chakraborty, R., 1991. Production of alkaline protease by a new *Aspergillus flavus* isolate under solid-substrate fermentation conditions for use as a depilation agent. *Appl. Environ. Microbiol.*, 57(3), pp.712-716.